Literature Survey on Multimodal and Agentic Retrieval-Augmented Generation Systems

1 Introduction

Retrieval-Augmented Generation (RAG) systems have emerged as an effective solution for reducing hallucination in Large Language Models (LLMs) by integrating external knowledge retrieval with generative reasoning. With the rise of Multimodal Large Language Models (MLLMs), modern RAG systems now support multiple modalities such as text, images, audio, video, and structured data. Recent research has further evolved toward graph-based and agentic RAG architectures capable of iterative planning, reasoning, and verification.

This literature survey analyzes major developments in Multimodal Retrieval-Augmented Generation (MM-RAG), graph-enhanced retrieval systems, and agentic frameworks. It highlights their contributions, limitations, and the improvements proposed in our project.

2 Literature Review

2.1 Comprehensive Survey on Multimodal RAG

Rui Zhang et al. presented a comprehensive survey on Multimodal Retrieval-Augmented Generation systems covering almost all combinations of input and output modalities including text, image, audio, video, code, and structured data. The paper introduced a taxonomy for MM-RAG systems and categorized retrieval and generation approaches across multiple modalities.

The survey identified four major stages of MM-RAG systems:

- Pre-retrieval
- Retrieval
- Augmentation
- Generation

The work also explored multiple multimodal applications such as:

- Visual Question Answering (VQA)

- Image Captioning
- Audio Captioning
- Video Question Answering
- Text-to-Image Generation
- Code Generation

2.1.1 Limitations

Despite its extensive coverage, the survey highlights several limitations in current MM-RAG systems:

- High computational cost due to multimodal encoding and retrieval.
- Difficulty in aligning heterogeneous semantic spaces across modalities.
- Large latency introduced by video and image processing pipelines.
- Heavy dependence on large-scale pretrained multimodal models.
- Lack of efficient reasoning mechanisms for dynamic retrieval.
- Limited exploration of many modality combinations.

2.1.2 Improvement Proposed in Our Work

Our proposed system improves upon these limitations by introducing:

- Adaptive retrieval mechanisms that dynamically decide when retrieval is necessary.
- Lightweight multimodal retrieval pipelines to reduce computational overhead.
- Confidence-aware retrieval routing for efficient modality selection.
- Agent-based iterative reasoning and verification loops.
- Hybrid graph-vector retrieval for better semantic and relational understanding.
- Explainable retrieval traces for transparency and debugging.

3 Scaling Beyond Context: MM-RAG for Document Understanding

Sensen Gao et al. explored Multimodal Retrieval-Augmented Generation systems specifically for document understanding. The paper focused on visually rich documents containing:

- Text
- Tables
- Charts
- Images
- Layout structures

The study classified MM-RAG systems based on:

- Open-domain and closed-domain retrieval
- Retrieval modality
- Retrieval granularity
- Graph-based enhancement
- Agent-based enhancement

The paper also reviewed multiple advanced frameworks such as:

- HM-RAG
- ViDoRAG
- mKG-RAG
- RECON
- LAD-RAG

3.1 Limitations

Although these systems improve document understanding, several critical limitations remain:

- Large visual token requirements exceed practical context limits.
- Heavy OCR dependence increases preprocessing complexity.
- Graph-based systems introduce storage and traversal overhead.
- Multi-agent systems suffer from coordination complexity.
- Retrieval at page-level granularity often misses fine-grained evidence.
- Long-document reasoning remains computationally expensive.

3.2 Improvement Proposed in Our Work

Our project addresses these issues through:

- Fine-grained region-level retrieval instead of only page-level retrieval.
- Dynamic graph generation instead of static knowledge graphs.
- Hierarchical agent collaboration with lightweight orchestration.
- Selective multimodal processing to reduce unnecessary computation.
- Memory-efficient retrieval compression techniques.
- Cross-modal verification for hallucination reduction.

4 MegaRAG: Multi-Modal Retrieval with Knowledge Graphs

MegaRAG introduced a multimodal retrieval framework combining visual and textual retrieval using knowledge graphs. The architecture enhanced semantic understanding through graph-based entity relationships and multimodal embeddings.

The system demonstrated improved contextual reasoning and stronger cross-modal retrieval performance compared to traditional retrieval systems.

4.1 Limitations

The major limitations of MegaRAG include:

- Heavy dependence on static knowledge graphs.
- High implementation and maintenance cost.
- Complex graph traversal operations increase latency.
- Scalability challenges for large datasets.
- Difficulty updating knowledge graphs dynamically.

4.2 Improvement Proposed in Our Work

Our proposed improvements include:

- Dynamic on-the-fly graph construction.
- Hybrid vector + graph retrieval architecture.
- Lightweight graph reasoning for scalability.
- Confidence-aware graph activation.
- Agentic retrieval refinement and verification.

5 Agentic Retrieval-Augmented Generation Systems

Recent research in Agentic RAG introduces autonomous agents capable of:

- Planning
- Retrieval orchestration
- Verification
- Reflection
- Multi-step reasoning

Systems such as HM-RAG and ViDoRAG employ multiple specialized agents for retrieval and reasoning tasks.

5.1 Limitations

Although agentic systems improve reasoning quality, they introduce several challenges:

- High inference latency
- Excessive token usage
- Multi-agent coordination overhead
- Increased architectural complexity
- High deployment cost

5.2 Improvement Proposed in Our Work

Our system improves agentic RAG through:

- Lightweight modular agents
- Adaptive reasoning depth
- Retrieval-triggered planning
- Self-reflective verification loops
- Efficient task decomposition

6 Proposed System

The proposed system is an **Adaptive Agentic Multimodal Graph-RAG Framework** designed to overcome the limitations of existing MM-RAG architectures.

6.1 Key Features

- Multimodal retrieval across text, images, tables, and documents
- Hybrid graph-vector retrieval
- Dynamic graph construction
- Fine-grained region-level evidence retrieval
- Agent-based reasoning and verification
- Confidence-aware retrieval routing
- Explainable reasoning traces
- Lightweight scalable architecture

6.2 Advantages Over Existing Systems

Existing Limitation	Existing Systems	Our Improvement
High computational overhead	MM-RAG systems	Adaptive selective retrieval
Static knowledge graphs	MegaRAG	Dynamic graph generation
Poor scalability	Graph-RAG systems	Lightweight graph reasoning
Long inference latency	Agentic RAG	Confidence-aware planning
Weak fine-grained retrieval	Page-level retrieval systems	Region-level retrieval
Hallucination issues	Traditional RAG	Self-verification loops
Cross-modal misalignment	Multimodal systems	Hybrid modality alignment

7 Conclusion

Multimodal Retrieval-Augmented Generation systems have significantly improved the reasoning and retrieval capabilities of modern AI systems. However, current approaches still face major challenges related to scalability, latency, graph complexity, multimodal alignment, and computational cost.

Our proposed Adaptive Agentic Multimodal Graph-RAG framework addresses these limitations through dynamic retrieval, lightweight graph reasoning, fine-grained evidence extraction, and self-reflective agentic workflows. The proposed architecture aims to provide a scalable, explainable, and computationally efficient solution for next-generation multimodal AI systems.

8 References

1. Rui Zhang et al., “A Comprehensive Survey on Multimodal RAG: All Combinations of Modalities as Input and Output”, 2025.

2. Sensen Gao et al., “Scaling Beyond Context: A Survey of Multimodal Retrieval-Augmented Generation for Document Understanding”, 2026.